

Statement of Purpose - Amazon ML Summer School 2026

My journey in AI and Machine Learning began with a curiosity about how intelligent systems learn from data and make decisions. As a B.Tech student specializing in Artificial Intelligence and Machine Learning, I have actively explored machine learning concepts through coursework, certifications, projects, and hands-on experimentation.

One of the first machine learning projects I built was a Boston House Price Prediction system using Python and Scikit-Learn. Through this project, I learned the complete workflow of a machine learning application, including data preprocessing, exploratory data analysis, feature engineering, model training, and evaluation using Linear Regression. Building this project helped me understand how mathematical concepts and data-driven approaches can be applied to solve real-world prediction problems.

To expand my understanding of AI beyond predictive modeling, I developed an AI-Integrated Hospital Portal with an NLP-based chatbot. While working on this project, I explored Natural Language Processing concepts and learned how AI can improve user interaction and accessibility. I also participated in a 24-hour Rapid Development Hackathon, where my team developed an AI-based healthcare solution and reached the semi-final stage. These experiences strengthened my problem-solving skills, teamwork, and ability to rapidly transform ideas into working prototypes.

Alongside projects, I pursued industry certifications including Oracle Certified Generative AI Professional and Salesforce Certified Agentforce Specialist. These certifications exposed me to modern AI systems and reinforced my interest in machine learning and intelligent applications.

Although I have gained practical exposure to AI and ML, I recognize several gaps in my knowledge. I want to deepen my understanding of the mathematical foundations of machine learning, particularly probability, statistics, linear algebra, optimization, and model generalization. I am also eager to learn how large-scale machine learning systems are designed, trained, evaluated, and deployed in real-world environments. While I have built projects using existing tools and frameworks, I want to better understand the reasoning behind algorithm selection, model behavior, and performance trade-offs.

Amazon ML Summer School is the ideal opportunity to bridge these gaps. Learning directly from Amazon Scientists and engaging with a structured curriculum will help me strengthen both my theoretical foundation and practical understanding of machine learning. I am particularly excited about interacting with peers who share similar interests and learning how industry experts approach complex ML problems.

I believe my combination of academic preparation, hands-on projects, certifications, hackathon experience, and genuine curiosity about machine learning makes me a strong fit for this program. More importantly, I have a strong desire to learn, continuously improve, and apply machine learning to solve meaningful real-world problems. I am confident that Amazon ML Summer School will significantly contribute to my growth as a future machine learning engineer and researcher.